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Prescribed finite-window reaction-time modality by conserved-budget support design

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E-mail: xiaoxiao.zhouyi@bristol.ac.uk**Keywords:** encounter reaction, reaction-time density, multimodality, conserved reactivity, Doi model

Abstract

Can a fixed amount of spatial reactivity be arranged to prescribe the finite-window topology of an encounter-reaction-time density? We answer this existence question for a Doi encounter process. For every fixed finite physical dimension $d \geq 2$, prescribed fixed finite integer m , and compact family of interior catalyst weights, an m -dependent design of ordered narrow catalyst slabs along a monotone exposure trajectory produces exactly m nondegenerate maxima and $m - 1$ nondegenerate minima on a declared compact positive-time window. A global zero bound for the separated Gaussian exposure clocks, a whole-window complement certificate, and compact-positive-time mixed-jet convergence transfer the complete stationary signature to the positive-budget killed process. The construction first fixes a sufficiently small noise-and-width scale and only then takes a sufficiently small installed reactivity budget. It is deliberately nonuniform: the geometry and admissible scales may depend on d and m , and no useful budget threshold, event-mass floor, or stationary-point count outside the declared window is obtained. The theorem therefore establishes prescribed finite modality under a conserved spatial resource without asserting a particular finite-parameter two-dimensional realization.

Draft compiled: August 7, 2026

1 Introduction

First-passage and reaction-time densities may develop several peaks when distinct trajectory families reach a reactive region on separated time scales [1, 2]. Continuum trap placement can alter capture-time topology [3], and two-particle encounter laws in physical dimensions two and three show related multirate mechanisms [4, 5, 6]. Spatially heterogeneous absorption and fixed-total-reactivity optimization are also well established [7, 8, 9]. Here we ask a narrower existence question: can the finite-window timing topology of a two-particle reaction be prescribed by support design while the installed amount of static reactivity remains fixed?

The fact that a univariate mixture of m Gaussian components has at most m modes is classical [10, 11]; that upper bound is not our contribution. The issue for an encounter process is to construct every stationary point with a strict margin, exclude additional stationary points throughout a declared time window after the contact factor is included, and carry the complete signature from free exposure to a killed Doi law.

We prove that this can be done for every prescribed fixed finite m in every fixed finite dimension $d \geq 2$. The supports, contact geometry, transport, initial law, and total installed centre-space reactivity are fixed within each constructed family. All interior allocations in that family share the same alternating signature. Thus the result concerns support-geometry design under a conserved budget; it is not a statement that one fixed support family switches among different mode counts when its weights are reallocated.

The Doi killing representation, the associated reactivity-times-occupancy identity, and Gaussian-mixture mode bounds are prior ingredients [12, 13, 14, 15]. The new content is their margin-bearing combination: a separated-clock construction with a global zero count, a uniform whole-window sign certificate, stability under the slowly varying contact probability, and a positive-budget transfer of the full finite stationary signature. The result concerns a transient density

under static spatial allocation and is distinct from universal mean-first-passage response to rare activation [16].

This article is self-contained. Sections 2–3 fix the model and the positive-time transfer estimate, section 4 states the modality theorem and explains its mechanism, and sections 5–6 set out the scope. The analytic realizations of the generator, the mixed-jet bound with its explicit constant, and the complete proof of the theorem are given in appendices A–C.

2 Conserved-reactivity encounter model

Let $q(x, t)$ be the unreacted configuration density of two mobile particles, with conservative generator $\mathcal{L}$. Write r for the relative coordinate and c for an encounter-centre coordinate. For a fixed contact profile $\chi_a(r)$ and normalized nonnegative centre-space profiles $\phi_j(c)$, define

$$\begin{aligned} \kappa_{\mathbf{w}}(c) &= B \sum_{j=1}^J w_j \phi_j(c), & w_j &\geq 0, \\ \sum_j w_j &= 1, & \int \phi_j(c) dc &= 1, \\ K_{B, \mathbf{w}}(x) &= \chi_a(r) \kappa_{\mathbf{w}}(c). \end{aligned} \tag{1}$$

Every admissible allocation has the same installed centre-space resource,

$$\int \kappa_{\mathbf{w}}(c) dc = B. \tag{2}$$

This is not an integral over the complete two-particle configuration space. The killed evolution, reaction-time density, and survival probability are

$$\begin{aligned} \partial_t q_{B, \mathbf{w}} &= (\mathcal{L} - K_{B, \mathbf{w}}) q_{B, \mathbf{w}}, & q_{B, \mathbf{w}}(0) &= q_0, \\ f_{B, \mathbf{w}}(t) &= \langle K_{B, \mathbf{w}}, q_{B, \mathbf{w}}(t) \rangle, & -S'_{B, \mathbf{w}}(t) &= f_{B, \mathbf{w}}(t). \end{aligned} \tag{3}$$

Equation (3) is the usual mass balance for a Doi volume-reaction process, not a novelty claim.

For longitudinal catalyst slabs, translation invariance of the common transverse coordinate gives an exact quotient

$$\Omega_\infty = \mathbb{R}_z \times \mathbb{R}_{r_\parallel} \times \mathbb{T}_W^{d-1}. \tag{4}$$

In the construction below the midpoint Z_t moves monotonically in mean, the relative process remains a positive distance inside the contact region along its deterministic trajectory on the chosen window, and each normalized slab is centred at the midpoint mean at a target time. The $W^{-(d-1)}$ factor in the killing field accounts for the omitted common transverse directions and preserves Eq. (2). Appendix A records the bounded reflecting/periodic and unbounded natural-decay realizations of this quotient together with the analyticity and positivity properties of the killed semigroup that the transfer estimate uses.

3 Positive-time transfer

Write $K_{B, \mathbf{w}} = BV_{\mathbf{w}}$ and introduce interior affine control coordinates $\mathbf{w}(\boldsymbol{\theta}) = \bar{\mathbf{w}} + P\boldsymbol{\theta}$. The normalized reaction density and the free-exposure law are

$$F_B(t, \boldsymbol{\theta}) = B^{-1} f_{B, \mathbf{w}(\boldsymbol{\theta})}(t), \quad G(t, \boldsymbol{\theta}) = \langle V_{\mathbf{w}(\boldsymbol{\theta})}, e^{t\mathcal{L}} q_0 \rangle. \tag{5}$$

On either a bounded reflecting/periodic quotient or the reversible natural-decay realization of Eq. (4), assume q_0 belongs to the corresponding state space and $V_{\mathbf{w}}$ is bounded on a complex control neighbourhood of a compact interior control set. A Dyson expansion followed by time and control Cauchy estimates gives, for every fixed time order r and finite control multi-index α ,

$$\sup_{(t, \boldsymbol{\theta}) \in [\tau, T] \times \Theta} |\partial_t^r \partial_{\boldsymbol{\theta}}^\alpha (F_B - G)| \leq C_{r, \alpha, \tau, T, \Theta} B. \tag{6}$$

Appendix B gives the explicit constant and the proof. In particular, C^2 convergence on a compact positive-time window preserves disjoint stationary-point neighbourhoods, curvature signs, endpoint slopes, and a nonzero derivative margin on their complement. The estimate is local in positive time and provides neither a useful numerical value of B nor control on the scale $t = O(B^{-1})$.

4 Prescribed finite modality from separated exposure clocks

The conserved-budget support construction admits encounter processes with any prescribed *fixed finite* number of reaction-time modes. We state this existence result in an exact quotient of a two-particle Doi process; the construction depends on the prescribed dimension and mode count and does not assert mode-count switching by weight redistribution on one fixed support.

Fix finite $d \geq 2$, finite $m \geq 1$, a transverse period $W > 0$, a contact radius $0 < a < W/2$, positive scales $\ell_0, \rho > 0$, and a compact interval $I = [\tau, T] \subset (0, \infty)$. On $\mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}$, consider the independent midpoint and relative processes

$$\begin{aligned} dZ_t &= -\gamma(Z_t - \bar{z}) dt + \varepsilon \sqrt{D_0} dW_t, \\ dR_{\parallel,t} &= -\gamma R_{\parallel,t} dt + 2\varepsilon \sqrt{D_0} dW_{\parallel,t}, \\ dR_{\perp,t} &= 2\varepsilon \sqrt{D_0} dW_{\perp,t} \pmod{W}. \end{aligned} \quad (7)$$

The mutually independent initial laws satisfy

$$\begin{aligned} Z_0 &\sim N(z_0, \varepsilon^2 D_0 / (2\gamma)), \\ R_{\parallel,0} &\sim N(r_{\parallel,0}, \varepsilon^2 u_0^2), \\ R_{\perp,0} &\sim \text{WN}(r_{\perp,0}, \varepsilon^2 \Sigma_{\perp,0}), \end{aligned} \quad \begin{aligned} D_0, \gamma, u_0 &> 0, \quad z_0 \neq \bar{z}, \\ u_0^2 &< 4D_0/\gamma, \quad \Sigma_{\perp,0} \succ 0. \end{aligned} \quad (8)$$

Here WN is the wrapped Gaussian on $\mathbb{T}_W^{d-1}$, $\Sigma_{\perp,0}$ is a positive-definite $(d-1) \times (d-1)$ covariance matrix, and $|r|_{\text{mi}}$ is the Euclidean norm of the longitudinal relative coordinate together with the shortest transverse representatives modulo W .

All driving Brownian motions are also independent. Thus $\mu(t) = \bar{z} + (z_0 - \bar{z})e^{-\gamma t}$ is strictly monotone and $\text{Var } Z_t = \varepsilon^2 D_0 / (2\gamma)$. Under the minimum-image convention, assume the deterministic relative trajectory $r_*(t) = (r_{\parallel,0}e^{-\gamma t}, r_{\perp,0})$ obeys $\sup_{t \in I} |r_*(t)|_{\text{mi}} \leq a - \eta$ for some $\eta > 0$.

Choose target times

$$\begin{aligned} \tau &< t_1 < \dots < t_m < T, \quad c_j = x(t_j), \\ x(t) &= \frac{\text{sgn}(\mu')\mu(t)}{\ell_0}, \end{aligned} \quad (9)$$

so that $x'(t)$ is bounded away from zero. At the corresponding physical centres, place normalized Gaussian catalyst slabs of width $\varepsilon\rho$. Their nonnegative weights belong to a compact subset of the simplex,

$$\mathcal{W}_{w_*} = \left\{ \mathbf{w} : \sum_{j=1}^m w_j = 1, w_j \geq w_* > 0 \right\}, \quad 0 < w_* \leq \frac{1}{m}, \quad (10)$$

and define

$$\begin{aligned} \phi_{j,\varepsilon}(z) &= \frac{\exp[-(z - \mu(t_j))^2 / (2\varepsilon^2 \rho^2)]}{\sqrt{2\pi} \varepsilon \rho}, \\ K_{B,\mathbf{w},\varepsilon}(Z, R) &= \mathbf{1}_{\{|R|_{\text{mi}} < a\}} \frac{B}{W^{d-1}} \sum_{j=1}^m w_j \phi_{j,\varepsilon}(Z). \end{aligned} \quad (11)$$

The slab normalization makes B the same installed centre-space budget for every allocation. Set $K_{B,\mathbf{w},\varepsilon} = B V_{\mathbf{w},\varepsilon}$, and let $T_0(t)$ and q_0 denote the unkilld semigroup and initial law. Then

$$\begin{aligned} G_{\varepsilon,\mathbf{w}}(t) &= B^{-1} \mathbb{E}_0[K_{B,\mathbf{w},\varepsilon}(Z_t, R_t)] = \langle V_{\mathbf{w},\varepsilon}, T_0(t) q_0 \rangle, \\ f_{B,\varepsilon,\mathbf{w}}(t) &= \mathbb{E}_0 \left[K_{B,\mathbf{w},\varepsilon}(Z_t, R_t) e^{-\int_0^t K_{B,\mathbf{w},\varepsilon}(Z_s, R_s) ds} \right]. \end{aligned} \quad (12)$$

The expectation $\mathbb{E}_0$ is with respect to the unkilld process started from Eq. (8). Independence gives the exact factorization

$$\begin{aligned} G_{\varepsilon,\mathbf{w}}(t) &= A_{\varepsilon}(t) H_{\sigma,\mathbf{w}}(x(t)), \\ A_{\varepsilon}(t) &= \frac{c_{d,\varepsilon}(t)}{W^{d-1} \sqrt{2\pi} \varepsilon \sqrt{D_0 / (2\gamma)} + \rho^2}, \end{aligned} \quad (13)$$

where $c_{d,\varepsilon}(t) = \mathbb{P}_0(|R_t|_{\text{mi}} < a)$ and $H_{\sigma,\mathbf{w}}$ is defined below.

Theorem 1 (Exact prescribed finite modality). *Fix finite $d \geq 2$, finite $m \geq 1$, the data above, and the compact window I . There is $\varepsilon_0 > 0$ such that, for every $0 < \varepsilon < \varepsilon_0$ and every $\mathbf{w} \in \mathcal{W}_{w_*}$, the unit-budget free-exposure clock has exactly m nondegenerate maxima and $m - 1$ nondegenerate minima on I , ordered alternately, and no stationary endpoint. For each fixed such ε , there is $B_0(\varepsilon) > 0$, uniform over $\mathcal{W}_{w_*}$, such that the budget-rescaled Doi reaction-time density $f_{B,\varepsilon,\mathbf{w}}/B$ has the same complete stationary signature for every $0 < B < B_0(\varepsilon)$. Multiplication by B leaves the stationary points unchanged, so the conclusion also holds for the physical density $f_{B,\varepsilon,\mathbf{w}}$.*

Theorem 1 is proved in appendix C, where it is restated with its complete hypothesis list as theorem C.8. The remainder of this section explains the mechanism.

The mechanism can be seen without invoking a normal form. Gaussian convolution reduces the longitudinal part of the unit-budget exposure to

$$H_{\sigma,\mathbf{w}}(x) = \sum_{j=1}^m w_j \exp\left[-\frac{(x - c_j)^2}{2\sigma^2}\right], \quad (14)$$

$$\sigma = \frac{\varepsilon}{\ell_0} \sqrt{\frac{D_0}{2\gamma} + \rho^2}.$$

Writing $q_j(x) = w_j \exp[-(x - c_j)^2/(2\sigma^2)]$, $\pi_j(x) = q_j(x)/H_{\sigma,\mathbf{w}}(x)$, and $L = \partial_x \log H$, let $\bar{c}(x)$ and $\text{Var}_{\pi(x)}(c)$ denote the mean and variance of the centres under the finite probability vector $\boldsymbol{\pi}(x)$. Then

$$L(x) = \frac{\bar{c}(x) - x}{\sigma^2}, \quad L'(x) = \frac{\text{Var}_{\pi(x)}(c)}{\sigma^4} - \frac{1}{\sigma^2}. \quad (15)$$

The univariate upper bound of m modes for an m -component Gaussian mixture is classical [10, 11]. Here a direct affine-exponential proof is retained because the Doi transfer needs all critical zeros counted with multiplicity together with uniform root-tube, complement, curvature, and endpoint margins.

For the adjacent pair $c_j < c_{j+1}$, the weighted crossover is

$$s_j = \frac{c_j + c_{j+1}}{2} + \frac{\sigma^2}{c_{j+1} - c_j} \log \frac{w_j}{w_{j+1}}. \quad (16)$$

The adjacent ratio obeys $q_{j+1}/q_j = 1/9$ at $s_j - \sigma^2 \log(9)/(c_{j+1} - c_j)$ and $q_{j+1}/q_j = 9$ at the corresponding plus edge. Thus both adjacent components remain uniformly present throughout the crossover, while all nonadjacent components are exponentially small. Equation (15) then yields one simple minimum at

$$r_j = s_j + O(\sigma^4), \quad L'(r_j) = \frac{(c_{j+1} - c_j)^2}{4\sigma^4} + O(\sigma^{-2}). \quad (17)$$

Each centre similarly contributes one simple maximum at $p_j = c_j + O(\exp[-q/\sigma^2])$, with $L'(p_j) = -\sigma^{-2} + o(\sigma^{-2})$. These constructions give $2m - 1$ simple stationary points. They are exhaustive: after multiplication by a positive Gaussian factor, H' is an exponential polynomial

$$\sum_{j=1}^m (a_j + b_j x) e^{\lambda_j x}, \quad \lambda_1 < \dots < \lambda_m, \quad (18)$$

which has at most $2m - 1$ real zeros counted with multiplicity. The extreme affine-exponential terms dominate in the two tails, so all zeros lie in a compact interval and real analyticity makes their number finite. A generalized Rolle induction then proves the bound by differentiating twice to remove the lowest-exponent affine term.

The remaining relative-coordinate contact probability is a positive slow factor. Uniform bounds on its first two logarithmic time derivatives, peak boxes of width $O(\sigma^2)$, valley boxes of width $O(\sigma^4)$, and a posterior-sector sign certificate exclude stationary points on the full box complement. Consequently the slow factor moves maxima by $O(\sigma^2)$ and minima by $O(\sigma^4)$ without changing their number or type. Finally, on every fixed positive ε , compact-positive-time mixed-jet convergence

$$\sup_{\mathbf{w} \in \mathcal{W}_{w_*}} \left\| \frac{f_{B,\varepsilon,\mathbf{w}}}{B} - G_{\varepsilon,\mathbf{w}} \right\|_{C^2(I)} \longrightarrow 0 \quad (B \downarrow 0) \quad (19)$$

preserves the disjoint root tubes, signed curvature margins, complement margin, and endpoint slopes.

The quantifier order is essential: the geometry and admissible scales may depend on fixed (d, m) ; one first fixes a sufficiently small $\varepsilon > 0$ and only then chooses $B < B_0(\varepsilon)$. The theorem does not provide a useful uniform lower bound for B_0 , a uniform-in- d or uniform-in- m construction, an event-mass floor, or a stationary-point count outside I ; it also uses a whole-window contact-interior condition whose small-noise limit is asymptotically saturated contact. The proof permits $B_0(\varepsilon)$ to shrink rapidly as ε decreases—potentially exponentially in $1/\varepsilon^2$ —and no rate is claimed; this is why no common useful budget is asserted across the family. Nontrivial contact, observability at a common positive budget, and finite-parameter robustness remain separate numerical validation gates.

5 Finite-parameter scope

The theorem is complete without a discretized example. Its proof establishes an exact existence statement for the continuum Doi process: it uses no finite-volume approximation, and no discretized convergence statement is needed for its conclusion. What the sequential small-parameter construction does not do is identify a practically resolved physical parameter set.

Accordingly, this article includes no numerical evidence for a particular finite-parameter realization, no off-lattice validation, no phase diagram, and no finite-parameter dimensional comparison. Its scientific conclusions rest only on the exact theorem.

We also make no rigorous numerical claim that a particular finite-volume scheme converges, together with its stationary roots, from bounded grids to the unbounded continuum process. Such a stronger numerical statement would require quantitative consistency, spatial and box-truncation error bounds, and componentwise margin-preserving transfer of each stationary component. Those additional ingredients are unnecessary for the theorem proved here and do not limit its scope.

6 Discussion

The construction turns catalyst locations into separated exposure clocks. Narrow Gaussian convolution creates one peak near each target time; an affine-exponential zero bound makes the count exhaustive; posterior-sector estimates control the complement after multiplication by the contact probability; and the positive-time estimate Eq. (6) transfers the strict topology to the Doi law. Together these steps establish arbitrary prescribed fixed finite modality under a conserved centre-space reactivity budget.

The restrictions are substantive. Geometry and admissible scales depend on d and m . The deterministic relative trajectory stays strictly inside the contact set on the design window, and its free contact probability approaches one there as the noise scale vanishes. The limits are sequential, observability at a common useful budget is not guaranteed, and nothing is proved about topology outside the declared window. These limits distinguish the exact existence theorem from a finite-parameter design rule while leaving its finite-window conclusion intact.

Data availability

This is a purely mathematical work. No new data were created or analysed in this study, and therefore no data-availability statement beyond this one is applicable.

Appendix A Doi model and analytic realizations

For a fixed finite physical dimension $d \geq 2$, the longitudinal-slab symmetry reduction gives

$$\mathcal{Q}_d^{\text{box}} = I_z \times I_{\parallel} \times \mathbb{T}_W^{d-1}, \quad \mathcal{Q}_d^{\infty} = \mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}. \quad (\text{A.1})$$

With $x = (z, r_{\parallel}, r_{\perp})$, equal particle diffusivities give the free forward operator

$$\mathcal{L}q = -\nabla \cdot (bq) + \nabla \cdot (D \nabla q), \quad D = \text{diag}(D/2, 2D, \dots, 2D), \quad (\text{A.2})$$

where

$$b(x) = (-\gamma(z - \bar{z}), -\gamma r_{\parallel}, 0, \dots, 0). \quad (\text{A.3})$$

The bounded quotient has zero forward flux on its nonperiodic faces and periodic transverse coordinates. Natural decay on the unbounded quotient is specified by the reversible density

$$\pi(x) = Z_{\pi}^{-1} \exp \left[-\frac{\gamma(z - \bar{z})^2}{D} - \frac{\gamma r_{\parallel}^2}{4D} \right], \quad (\text{A.4})$$

including the normalized uniform torus factor.

Let $0 < a < W/2$, let $\chi_a(r)$ be the embedded minimum-image contact-ball indicator, and let normalized nonnegative longitudinal profiles obey $\phi_j \in L^\infty$ and $\int \phi_j(z) dz = 1$. Put

$$V_j(z, r) = W^{-(d-1)} \chi_a(r) \phi_j(z), \quad V_w = \sum_{j=1}^J w_j V_j, \quad K_{B,w} = BV_w, \quad (\text{A.5})$$

where $w_j \geq 0$ and $\sum_j w_j = 1$. The installed centre-space catalyst amount is B for every w . The killed state and normalized reaction density are

$$q_{B,w}(t) = e^{t(\mathcal{L} - BM_{V_w})} q_0, \quad F_B(t, w) = \langle V_w, q_{B,w}(t) \rangle. \quad (\text{A.6})$$

Proposition A.1 (Analytic killed semigroups). *On $\mathcal{Q}_d^{\text{box}}$, the reflecting/periodic realization of $\mathcal{L}$ on $L^2(\mathcal{Q}_d^{\text{box}})$ generates a positive analytic semigroup. On $\mathcal{Q}_d^\infty$, the natural-decay realization on $X_\pi = L^2(\pi^{-1}dx)$ does the same. For real $B \geq 0$ and simplex control w , bounded multiplication by $-BV_w$ preserves the generator domain and yields a positive, mass-decreasing analytic semigroup. At positive time the state is entire in every finite affine complexification of the control amplitudes. On $\mathcal{Q}_d^\infty$, the observable pairing is continuous when $V_w \in L^2(\pi dx) \cap L^\infty$.*

Proof. Set $q = \pi u$. The identity

$$\mathcal{L}(\pi u) = \pi \mathcal{G}u, \quad \mathcal{G} = \pi^{-1} \nabla \cdot (\pi \mathbf{D} \nabla) \quad (\text{A.7})$$

follows from $b = \mathbf{D} \nabla \log \pi$. On $L^2(\pi dx)$, the closed weighted form has

$$\langle u, \mathcal{G}v \rangle_{L^2(\pi)} = - \int (\nabla u)^\top \mathbf{D} \nabla v \pi dx. \quad (\text{A.8})$$

It therefore realizes $\mathcal{G}$ as a nonpositive self-adjoint generator. Multiplication by π is a bounded isomorphism on the bounded quotient and a unitary map into X_π on the unbounded quotient. Bounded multiplication by $-BV_w$ preserves analyticity and the domain. The norm-convergent Dyson–Phillips series gives entire dependence on the finite control amplitudes. Positivity and mass decrease for real nonnegative data follow from the product formula, equivalently from Feynman–Kac [15, 17]. $\square$

Appendix B Uniform compact-positive-time mixed derivatives

Choose an interior affine control chart $w(\theta) = w_* + P\theta$ and a compact real control set Θ . Fix $\delta > 0$ such that each closed coordinate polydisc of radius δ about a point of Θ lies in the complex tube

$$\Theta_\delta^+ = \{\zeta \in \mathbb{C}^{J-1} : \text{dist}_\infty(\zeta, \Theta) < 2\delta\}. \quad (\text{B.1})$$

Define

$$G(t, \theta) = \langle V_{w(\theta)}, e^{t\mathcal{L}} q_0 \rangle, \quad v_{\infty, \delta} = \sup_{\zeta \in \Theta_\delta^+} \|V_{w(\zeta)}\|_\infty. \quad (\text{B.2})$$

Let $v_{*, \delta}$ be the supremum of $\|V_{w(\zeta)}\|_{L^2}$ on the bounded quotient and of $\|V_{w(\zeta)}\|_{L^2(\pi dx)}$ on the unbounded quotient. Write κ_π for the bounded-quotient similarity bound and set $\kappa_\pi = 1$ on the unbounded quotient.

Theorem B.1 (Uniform weak-reaction mixed-jet bridge). *Fix $0 < \tau \leq T < \infty$, $0 \leq B \leq B_{\max}$, an integer $r \geq 0$, and a finite control multi-index α . The normalized density $F_B(t, \theta)$ in Eq. (A.6) extends continuously to $F_0 = G$, and*

$$\begin{aligned} & \sup_{(t, \theta) \in [\tau, T] \times \Theta} |\partial_t^r \partial_\theta^\alpha (F_B - G)| \\ & \leq r! \alpha! \left(\frac{2}{\tau}\right)^r \delta^{-|\alpha|} v_{*, \delta} \kappa_\pi \left[\exp\left(\frac{3}{2} B v_{\infty, \delta} T\right) - 1 \right] \|q_0\|_X. \end{aligned} \quad (\text{B.3})$$

In particular, the left side is at most $BC_{r, \alpha}$ for a finite constant independent of $B \in [0, B_{\max}]$. This is the explicit form of Eq. (6).

Proof. For complex time z with $\text{Re } z > 0$, the norm-convergent Dyson expansion in the self-adjoint representation is

$$e^{z(\mathcal{G} - BV_w)} = e^{z\mathcal{G}} + \sum_{n=1}^{\infty} (-Bz)^n \int_{\Delta_n} e^{z(1-s_1)\mathcal{G}} V_w \left[\prod_{k=1}^{n-1} e^{z(s_k - s_{k+1})\mathcal{G}} V_w \right] e^{zs_n \mathcal{G}} d\mathbf{s}, \quad (\text{B.4})$$

where $\Delta_n = \{1 \geq s_1 \geq \dots \geq s_n \geq 0\}$ has volume $1/n!$. Every free factor is contractive. Consequently,

$$\|e^{z(\mathcal{L}-BV_w)} - e^{z\mathcal{L}}\|_{X \rightarrow X} \leq \kappa_\pi \{e^{B|z|\|V_w\|_\infty} - 1\}. \quad (\text{B.5})$$

The complex-bilinear observable remainder thus satisfies

$$\left| \left\langle V_{w(\zeta)}, [e^{z(\mathcal{L}-BV_{w(\zeta)})} - e^{z\mathcal{L}}]q_0 \right\rangle \right| \leq v_{*,\delta} \kappa_\pi \{e^{Bv_{\infty,\delta}|z|} - 1\} \|q_0\|_X. \quad (\text{B.6})$$

For $t \in [\tau, T]$, the disk $|z - t| \leq \tau/2$ remains in the open right half-plane and has $|z| \leq 3T/2$. Cauchy's estimate on this disk and on each radius- δ control polydisc proves Eq. (B.3). Finally, $e^{Bx} - 1 \leq Bxe^{B_{\max}x}$ gives the linear bound. $\square$

The estimate transfers any fixed finite collection of strict stationary-point neighbourhood, curvature, endpoint, and complement inequalities. It is asserted only on compact positive-time intervals and does not supply a useful numerical budget threshold.

Appendix C Exact prescribed finite modality: complete proof

This appendix proves the exact finite-mode statement announced as theorem 1 in section 4. The construction is pointwise in a fixed finite dimension and a fixed finite mode count. Its two small parameters are ordered: first the narrow-noise parameter is fixed sufficiently small, and only then is the Doi budget taken sufficiently small. All stationary-point counts below are on the declared compact positive-time window.

C.1 Exact quotient, stationary midpoint variance, and whole-window contact

Fix finite integers $d \geq 2$ and $m \geq 1$, a transverse period $W > 0$, a contact radius $0 < a < W/2$, and a compact positive-time interval $I = [\tau, T] \subset (0, \infty)$. On $\mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}$, let the longitudinal midpoint and relative coordinates solve

$$dZ_t = -\gamma(Z_t - \bar{z})dt + \varepsilon\sqrt{D_0}dW_t, \quad (\text{C.1})$$

$$dR_{\parallel,t} = -\gamma R_{\parallel,t}dt + 2\varepsilon\sqrt{D_0}dW_{\parallel,t}, \quad (\text{C.2})$$

$$dR_{\perp,t} = 2\varepsilon\sqrt{D_0}dW_{\perp,t} \pmod{W}, \quad (\text{C.3})$$

where $D_0, \gamma > 0$. All Brownian drivers and all initial variables are mutually independent, and

$$Z_0 \sim N\left(z_0, \varepsilon^2 \frac{D_0}{2\gamma}\right), \quad z_0 \neq \bar{z}, \quad (\text{C.4})$$

$$R_{\parallel,0} \sim N(r_{\parallel,0}, \varepsilon^2 u_0^2), \quad 0 < u_0^2 < \frac{4D_0}{\gamma}, \quad (\text{C.5})$$

$$R_{\perp,0} \sim \text{WN}(r_{\perp,0}, \varepsilon^2 \Sigma_{\perp,0}), \quad \Sigma_{\perp,0} \succ 0. \quad (\text{C.6})$$

Here WN denotes the wrapped Gaussian on the transverse torus. The two strict variance inequalities $D_0/(2\gamma) < D_0/\gamma$ and $u_0^2 < 4D_0/\gamma$ are precisely the unbounded Gaussian conditions that place this initial law in the weighted state space used by theorem B.1; the wrapped factor is square integrable against the uniform torus measure.

The midpoint has mean and variance

$$\mu(t) = \bar{z} + (z_0 - \bar{z})e^{-\gamma t}, \quad \text{Var } Z_t = \varepsilon^2 \frac{D_0}{2\gamma} \quad (t \geq 0). \quad (\text{C.7})$$

Thus only the variance, not the mean, is stationary. This constant variance is what produces a common-variance Gaussian mixture below.

Let $\varsigma = \text{sgn } \mu' \in \{-1, 1\}$, fix a physical reference length $\ell_0 > 0$, and define

$$x(t) = \frac{\varsigma\mu(t)}{\ell_0}, \quad v_0 := \inf_{t \in I} x'(t) > 0. \quad (\text{C.8})$$

Choose target times and their dimensionless centres by

$$\tau < t_1 < \dots < t_m < T, \quad c_j = x(t_j). \quad (\text{C.9})$$

Then $c_1 < \dots < c_m$, and the physical centre of slab j is $\varsigma\ell_0 c_j = \mu(t_j)$. In particular, reversing the trajectory orientation reverses the physical centres at the same time; no centre is held fixed under the coordinate reversal.

Write

$$r_*(t) = (r_{\parallel,0}e^{-\gamma t}, r_{\perp,0}) \quad (\text{C.10})$$

and impose the whole-window, minimum-image contact margin

$$\sup_{t \in I} |r_*(t)|_{\text{mi}} \leq a - \eta \quad \text{for some } \eta > 0. \quad (\text{C.11})$$

The word “whole-window” is essential: the condition is required on every point of I , not merely near the target times.

Lemma C.1 (Whole-window differentiated contact tail). *Let*

$$c_{d,\varepsilon}(t) = \mathbb{P}\{|R_t|_{\text{mi}} < a\}. \quad (\text{C.12})$$

Under Eqs. (C.2)–(C.11), for $r = 0, 1, 2$ there are finite constants $C_{r,d}, N_{r,d}, q_d > 0$, independent of all sufficiently small ε , such that

$$\|\partial_t^r(c_{d,\varepsilon} - 1)\|_{L^\infty(I)} \leq C_{r,d}\varepsilon^{-N_{r,d}} \exp(-q_d/\varepsilon^2). \quad (\text{C.13})$$

Consequently, $c_{d,\varepsilon} \geq 1/2$ for sufficiently small ε , and the first two logarithmic derivatives of $c_{d,\varepsilon}$ are uniformly bounded on I and converge to zero faster than any power of ε .

Proof. In an unwrapped transverse chart the relative Gaussian has mean $r_*(t)$ and covariance $\varepsilon^2 \Sigma_R(t)$, with longitudinal and transverse blocks

$$u_0^2 e^{-2\gamma t} + \frac{2D_0}{\gamma}(1 - e^{-2\gamma t}), \quad \Sigma_{\perp,0} + 4D_0 t I_{d-1}. \quad (\text{C.14})$$

Because $I \subset (0, \infty)$, $u_0 > 0$, and $\Sigma_{\perp,0} \succ 0$, this covariance and its first two time derivatives are uniformly bounded on I , and its eigenvalues lie between two positive constants. The reverse triangle inequality and Eq. (C.11) give

$$\inf_{t \in I} \inf_{|r|_{\text{mi}} \geq a} d_{\text{cyl}}(r, r_*(t)) \geq \eta. \quad (\text{C.15})$$

The assumption $a < W/2$ keeps the contact ball away from the transverse cut locus. Express the wrapped density as its Gaussian lattice-image sum. Every time derivative of order at most two is the same Gaussian multiplied by a polynomial in the scaled displacement and in ε^{-1} , with degree bounded independently of t . Equation (C.15), the uniform covariance bounds, and a standard Gaussian tail estimate therefore give, after summing the lattice images,

$$\sup_{t \in I} \int_{|r|_{\text{mi}} \geq a} |\partial_t^r p_{\varepsilon,t}(r)| \, dr \leq C_{r,d} \varepsilon^{-N_{r,d}} e^{-q_d/\varepsilon^2}, \quad r = 0, 1, 2. \quad (\text{C.16})$$

Differentiation under the integral is justified by the same integrable majorant and yields Eq. (C.13). For small ε , its $r = 0$ case gives $c_{d,\varepsilon} \geq 1/2$, while

$$\begin{aligned} |\partial_t \log c_{d,\varepsilon}| &\leq 2|\partial_t c_{d,\varepsilon}|, \\ |\partial_t^2 \log c_{d,\varepsilon}| &\leq 2|\partial_t^2 c_{d,\varepsilon}| + 4|\partial_t c_{d,\varepsilon}|^2. \end{aligned} \quad (\text{C.17})$$

This proves the logarithmic-derivative statement. $\square$

Fix $\rho > 0$, put

$$S_*^2 = \frac{D_0}{2\gamma} + \rho^2, \quad \sigma = \frac{\varepsilon S_*}{\ell_0}, \quad (\text{C.18})$$

and use the normalized physical slabs

$$\phi_{j,\varepsilon}(z) = \frac{1}{\sqrt{2\pi} \varepsilon \rho} \exp\left[-\frac{(z - \mu(t_j))^2}{2\varepsilon^2 \rho^2}\right]. \quad (\text{C.19})$$

Let $0 < w_* \leq 1/m$ and

$$\mathcal{W}_{w_*} = \left\{ w = (w_1, \dots, w_m) : \sum_{j=1}^m w_j = 1, \quad w_j \geq w_* > 0 \right\}. \quad (\text{C.20})$$

For $w \in \mathcal{W}_{w_*}$, define the Doi killing field

$$K_{B,w,\varepsilon}(Z, R) = \mathbf{1}_{\{|R|_{\text{mi}} < a\}} \frac{B}{W^{d-1}} \sum_{j=1}^m w_j \phi_{j,\varepsilon}(Z) = BV_{w,\varepsilon}(Z, R). \quad (\text{C.21})$$

Every slab has unit full longitudinal integral, so B is the same installed centre-space budget for every allocation.

Lemma C.2 (Exact common-variance free-exposure factorization). *Let $T_0(t)$ be the unskilled semigroup and q_0 the initial law above. The unit-budget free-exposure clock is exactly*

$$G_{\varepsilon,w}(t) := \langle V_{w,\varepsilon}, T_0(t)q_0 \rangle = a_\varepsilon(t) H_{\sigma,w}(x(t)), \quad (\text{C.22})$$

$$a_\varepsilon(t) := \frac{c_{d,\varepsilon}(t)}{W^{d-1} \sqrt{2\pi} \varepsilon S_*}, \quad (\text{C.23})$$

$$H_{\sigma,w}(x) := \sum_{j=1}^m w_j \exp \left[-\frac{(x - c_j)^2}{2\sigma^2} \right]. \quad (\text{C.24})$$

The positive factor a_ε has uniformly bounded first two logarithmic time derivatives for all sufficiently small ε .

Proof. Independence factors the contact event from every longitudinal slab expectation. By Eq. (C.7), convolution of $\phi_{j,\varepsilon}$ with the law of Z_t gives

$$\mathbb{E}[\phi_{j,\varepsilon}(Z_t)] = \frac{1}{\sqrt{2\pi} \varepsilon S_*} \exp \left[-\frac{(\mu(t) - \mu(t_j))^2}{2\varepsilon^2 S_*^2} \right]. \quad (\text{C.25})$$

Since $\mu(t) - \mu(t_j) = \varsigma \ell_0(x(t) - c_j)$, Eqs. (C.18) and (C.25) give Eqs. (C.22)–(C.24). The last assertion follows from lemma C.1, because the remaining factors in a_ε are independent of time. $\square$

C.2 The pure common-variance mixture

Set $J = x(I) = [\alpha, \beta]$. The targets are interior, so

$$\alpha < c_1 < \dots < c_m < \beta. \quad (\text{C.26})$$

For $m \geq 2$, let

$$\Delta_j = c_{j+1} - c_j, \quad \Delta_* := \min_{1 \leq j < m} \Delta_j > 0. \quad (\text{C.27})$$

No empty minimum is used when $m = 1$.

Define

$$q_j(x) = w_j e^{-(x-c_j)^2/(2\sigma^2)}, \quad \pi_j(x) = \frac{q_j(x)}{H_{\sigma,w}(x)}, \quad \bar{c}(x) = \sum_{j=1}^m \pi_j(x) c_j. \quad (\text{C.28})$$

The logarithmic slope $L_{\sigma,w} = \partial_x \log H_{\sigma,w}$ obeys the exact identities

$$L_{\sigma,w}(x) = \frac{\bar{c}(x) - x}{\sigma^2}, \quad (\text{C.29})$$

$$L'_{\sigma,w}(x) = \frac{\text{Var}_{\pi(x)}(c)}{\sigma^4} - \frac{1}{\sigma^2}. \quad (\text{C.30})$$

Indeed, differentiation of q_j gives the first identity, while $\pi'_j = \pi_j(c_j - \bar{c})/\sigma^2$ gives the second.

The mode cap for univariate Gaussian mixtures is classical [10, 11]. We retain the direct multiplicity-counting proof because the later Doi transfer also needs explicit uniform root locations, curvature scales, and full-complement margins.

Lemma C.3 (Global extended-Chebyshev zero bound). *For distinct real centres and positive weights, $H'_{\sigma,w}$ has at most $2m - 1$ real zeros counted with multiplicity. Its real zero set is finite.*

Proof. Multiplication by a nowhere-zero function preserves both zeros and their multiplicities, and direct differentiation gives

$$\sigma^2 e^{x^2/(2\sigma^2)} H'_{\sigma,w}(x) = \sum_{j=1}^m w_j (c_j - x) e^{-c_j^2/(2\sigma^2)} e^{c_j x/\sigma^2}. \quad (\text{C.31})$$

We prove the slightly more general claim that a nonzero function

$$P_m(x) = \sum_{j=1}^m (a_j + b_j x) e^{\lambda_j x}, \quad \lambda_1 < \dots < \lambda_m, \quad (\text{C.32})$$

with no identically zero summand has at most $2m - 1$ real zeros counted with multiplicity.

First, the zero set is finite. After multiplication by $e^{-\lambda_1 x}$, the first affine term dominates as $x \rightarrow -\infty$; the last affine-exponential term dominates as $x \rightarrow +\infty$. Each dominating affine factor has a fixed sign sufficiently far in its tail. Thus all zeros lie in a compact interval. A nonzero real-analytic function has only finitely many zeros there.

Now induct on m . The $m = 1$ case is the affine zero bound. For $m > 1$, put $Q = e^{-\lambda_1 x} P_m$. Two derivatives annihilate the first affine term. For every $j \geq 2$, the second derivative of $(a_j + b_j x) e^{(\lambda_j - \lambda_1)x}$ is a nonzero affine polynomial times the same exponential, and the remaining positive exponents are distinct. If a C^2 function has $N \geq 2$ real zeros counted with multiplicity, generalized Rolle counting gives at least $N - 2$ zeros of its second derivative counted with multiplicity: differentiation reduces the multiplicity at each root, and ordinary Rolle supplies a root between each pair of distinct roots. Hence

$$N(Q) - 2 \leq N(Q'') \leq 2(m - 1) - 1, \quad (\text{C.33})$$

so $N(P_m) \leq 2m - 1$; the cases $N < 2$ are immediate. In Eq. (C.31), every affine coefficient has nonzero slope $-w_j e^{-c_j^2/(2\sigma^2)}$, so the claim applies. $\square$

Lemma C.4 (Uniform adjacent isolation). *For the fixed finite centre set and the compact weight family, there are $C_{\text{iso}}, q_{\text{iso}} > 0$ such that, for every $j < m$,*

$$\sup_{x \in [c_j, c_{j+1}]} \frac{\sum_{k \notin \{j, j+1\}} q_k(x)}{q_j(x) + q_{j+1}(x)} \leq C_{\text{iso}} e^{-q_{\text{iso}}/\sigma^2} \quad (\text{C.34})$$

for all sufficiently small σ , uniformly in w . On $[\alpha, c_1]$, component 1 has the analogous one-component bound, and on $[c_m, \beta]$, component m does.

Proof. For $x \in [c_j, c_{j+1}]$ and $k < j$,

$$(x - c_k)^2 - (x - c_j)^2 = (c_j - c_k)(2x - c_j - c_k) \geq (c_j - c_k)^2. \quad (\text{C.35})$$

For $k > j + 1$, comparison with c_{j+1} gives the corresponding positive lower bound. Since $w_k/w_j \leq 1/w_*$, summation over the finite centre set proves Eq. (C.34). The same difference-of-squares comparison proves the two tail statements. The posterior mean and variance of the full mixture therefore differ from those of the isolated adjacent pair by $O(e^{-q/\sigma^2})$, uniformly in the gap, the allocation, and sufficiently small σ . $\square$

Theorem C.5 (Exact topology of the pure mixture). *Uniformly for $w \in \mathcal{W}_{w_*}$, every sufficiently small $\sigma > 0$ gives exactly m simple local maxima and $m - 1$ simple local minima of $H_{\sigma, w}$ on J , ordered alternately, with nonzero endpoint derivatives. The maximum near c_j , denoted $p_j(\sigma, w)$, satisfies*

$$p_j = c_j + O(e^{-q/\sigma^2}), \quad L'_{\sigma, w}(p_j) = -\sigma^{-2} + o(\sigma^{-2}). \quad (\text{C.36})$$

For $m \geq 2$, define the weighted equality point

$$s_j(\sigma, w) = \frac{c_j + c_{j+1}}{2} + \frac{\sigma^2}{\Delta_j} \log \frac{w_j}{w_{j+1}}. \quad (\text{C.37})$$

The minimum in the j -th gap, denoted $r_j(\sigma, w)$, satisfies

$$r_j = s_j + O(\sigma^4), \quad (\text{C.38})$$

$$L'_{\sigma, w}(r_j) = \frac{\Delta_j^2}{4\sigma^4} + O(\sigma^{-2}). \quad (\text{C.39})$$

All remainders are uniform on the compact weight set.

Proof. Fix a constant $A > 1$. On $|x - c_j| \leq A\sigma^2$, one-component isolation gives

$$L_{\sigma,w}(x) = \frac{c_j - x}{\sigma^2} + O(\sigma^{-2}e^{-q/\sigma^2}), \quad (\text{C.40})$$

$$L'_{\sigma,w}(x) = -\sigma^{-2} + O(\sigma^{-4}e^{-q/\sigma^2}). \quad (\text{C.41})$$

At $c_j - \sigma^2$ the slope is positive, at $c_j + \sigma^2$ it is negative, and it is strictly decreasing between them. Hence there is one simple maximum with the location and derivative in Eq. (C.36).

For a gap, the isolated adjacent odds are exactly

$$\frac{q_{j+1}(x)}{q_j(x)} = \exp\left[\frac{\Delta_j(x - s_j)}{\sigma^2}\right]. \quad (\text{C.42})$$

Put $h_j = (\log 9)/\Delta_j$ and

$$C_j = [s_j - h_j\sigma^2, s_j + h_j\sigma^2]. \quad (\text{C.43})$$

At the left and right endpoints the adjacent posterior masses are, respectively, $(9/10, 1/10)$ and $(1/10, 9/10)$. By lemma C.4, after reducing a common σ_0 , both adjacent posterior masses are bounded below throughout C_j , and

$$\text{Var}_{\pi(x)}(c) \geq \kappa_0 \Delta_*^2, \quad L'_{\sigma,w}(x) \geq \frac{\kappa_v}{\sigma^4} > 0 \quad (x \in C_j) \quad (\text{C.44})$$

for constants independent of j, w, σ . At the two endpoints of C_j , Eq. (C.29) gives, for sufficiently small σ ,

$$L_{\sigma,w}(s_j - h_j\sigma^2) \leq -\frac{\Delta_j}{5\sigma^2}, \quad L_{\sigma,w}(s_j + h_j\sigma^2) \geq \frac{\Delta_j}{5\sigma^2}. \quad (\text{C.45})$$

Thus C_j contains exactly one simple minimum root.

At s_j , the adjacent posterior masses are equal and the adjacent posterior mean is $(c_j + c_{j+1})/2$. Hence

$$L_{\sigma,w}(s_j) = -\frac{1}{\Delta_j} \log \frac{w_j}{w_{j+1}} + O(\sigma^{-2}e^{-q/\sigma^2}) = O(1) \quad (\text{C.46})$$

uniformly in w . The mean-value theorem and Eq. (C.44) give $r_j - s_j = O(\sigma^4)$. At r_j , the two adjacent posterior masses are $1/2 + O(\sigma^2)$; substitution in Eq. (C.30) proves Eq. (C.39).

For small σ , the constructed peak and gap neighbourhoods are pairwise disjoint and contained in (α, β) . They contain $m + (m - 1) = 2m - 1$ distinct simple stationary points. The global bound in lemma C.3 excludes every additional real stationary point, including multiple ones. Tail isolation and the fixed separations $c_1 - \alpha > 0$ and $\beta - c_m > 0$ give

$$L_{\sigma,w}(\alpha) > 0, \quad L_{\sigma,w}(\beta) < 0, \quad (\text{C.47})$$

so neither endpoint is stationary. When $m = 1$, there are no gaps or valley definitions: the exact identity $L_{\sigma,w}(x) = (c_1 - x)/\sigma^2$ gives one maximum and both endpoint signs directly. $\square$

C.3 Full posterior-sector certificate and a slow positive factor

The zero bound above does not apply after multiplication by a nonconstant positive factor. We therefore certify the complete complement of small peak and valley boxes.

Lemma C.6 (Posterior-sector certificate). *Fix $K > 0$. There are constants*

$A_p, A_v, C_p, C_v, \kappa_p, \kappa_v > 0$ and $\sigma_K > 0$, depending only on the fixed centre set, J , w_ , and K , such that, for every $0 < \sigma < \sigma_K$ and every admissible w , the boxes*

$$P_j = [c_j - A_p\sigma^2, c_j + A_p\sigma^2], \quad (\text{C.48})$$

$$V_j = [r_j - A_v\sigma^4, r_j + A_v\sigma^4] \quad (j < m) \quad (\text{C.49})$$

are pairwise disjoint and contained in (α, β) , and the following hold:

$$|L_{\sigma,w}| \leq C_p, \quad L'_{\sigma,w} \leq -\kappa_p\sigma^{-2} \quad \text{on } P_j, \quad (\text{C.50})$$

$$|L_{\sigma,w}| \leq C_v, \quad L'_{\sigma,w} \geq \kappa_v\sigma^{-4} \quad \text{on } V_j. \quad (\text{C.51})$$

At the left and right box boundaries, respectively, the slope signs are

	left boundary	right boundary
P_j	$L_{\sigma,w} \geq 2K$	$L_{\sigma,w} \leq -2K$
V_j	$L_{\sigma,w} \leq -2K$	$L_{\sigma,w} \geq 2K$

(C.52)

On the entire complement in J of the open boxes,

$$|L_{\sigma,w}(x)| \geq 2K, \quad (\text{C.53})$$

with the exhaustive alternating sign order

$$+ P_1 - V_1 + P_2 - \cdots + P_m - . \quad (\text{C.54})$$

For $m = 1$, every valley clause is omitted.

Proof. Choose $A_p \geq 4K$. The uniform expansions (C.40)–(C.41) give, after reducing σ_K , Eq. (C.50) and the peak row of Eq. (C.52). The convex-hull property $\bar{c}(x) \in [c_1, c_m]$ gives the full outer-tail estimates without any local expansion:

$$\begin{aligned} x \leq c_1 - A_p \sigma^2 &\implies L_{\sigma,w}(x) \geq A_p, \\ x \geq c_m + A_p \sigma^2 &\implies L_{\sigma,w}(x) \leq -A_p. \end{aligned} \quad (\text{C.55})$$

Fix a gap and abbreviate $\Delta = \Delta_j$, $s = s_j$. The right adjacent posterior mass is

$$p(x) = \frac{q_{j+1}(x)}{q_j(x) + q_{j+1}(x)} = \frac{1}{1 + \exp[-\Delta(x - s)/\sigma^2]}. \quad (\text{C.56})$$

On the crossover interval C_j in Eq. (C.43), its edge odds are $1/9$ and 9 , not exponentially small or large. Both adjacent masses are therefore bounded below throughout C_j , and Eq. (C.44) holds there. Conversely, the fixed finite centre diameter bounds the posterior variance, so $|L'| \leq C\sigma^{-4}$ on C_j .

The pure root satisfies $L(r_j) = 0$. Choose A_v large enough that $\kappa_v A_v \geq 2K$. Since $A_v \sigma^4 = o(\sigma^2)$, the valley box lies in C_j for small σ . Integrating the lower and upper bounds for L' from r_j proves Eq. (C.51), the valley row of Eq. (C.52), and the required negative and positive bounds on $C_j \setminus \text{int } V_j$.

It remains to cover both outer sectors of every gap; this is the step that prevents an unproved dominance claim at a crossover edge. Consider first

$$[c_j + A_p \sigma^2, s_j - h_j \sigma^2] \quad (\text{C.57})$$

and split it at $c_j + \Delta_j/4$, which lies in the stated interval for all sufficiently small σ , uniformly in w . On the portion nearer c_j , every other component is exponentially small relative to component j , so

$$\bar{c}(x) - x \leq -\frac{1}{2}A_p \sigma^2, \quad L_{\sigma,w}(x) \leq -A_p/2 \leq -2K. \quad (\text{C.58})$$

On the remaining portion, monotonicity of Eq. (C.56) and the left crossover edge give $p(x) \leq 1/10$, while $x - c_j \geq \Delta_j/4$. The adjacent-pair mean then obeys

$$c_j + \Delta_j p(x) - x \leq \frac{\Delta_j}{10} - \frac{\Delta_j}{4} = -\frac{3\Delta_j}{20}. \quad (\text{C.59})$$

The nonadjacent correction is exponentially small by lemma C.4; hence this sector also has $L \leq -2K$ for small σ , including its $1/9$ -odds edge.

Symmetrically, split $[s_j + h_j \sigma^2, c_{j+1} - A_p \sigma^2]$ at $c_{j+1} - \Delta_j/4$. On the portion nearer the crossover, $p(x) \geq 9/10$ and $c_{j+1} - x \geq \Delta_j/4$, giving the positive analogue of Eq. (C.59); on the portion nearer c_{j+1} , one-component isolation gives $L \geq A_p/2 \geq 2K$. Thus the right outer sector has $L \geq 2K$.

The fixed centre and endpoint separations make all boxes disjoint and interior for one common sufficiently small σ_K . The outer tails, all peak boundaries, both outer sectors of every gap, and every crossover-minus-valley sector cover every point of the box complement. This proves Eqs. (C.53)–(C.54). When $m = 1$, the same result follows directly from $L = (c_1 - x)/\sigma^2$ and Eq. (C.55). $\square$

Theorem C.7 (Slow positive factors preserve the exact topology). *Let $x \in C^2(I)$ be strictly increasing with $\inf_I x' = v_0 > 0$. Let $a_\sigma \in C^2(I)$ be positive and assume*

$$M_0 := \sup_{\sigma < \sigma_0} \|\partial_t \log a_\sigma\|_{L^\infty(I)} < \infty, \quad M_1 := \sup_{\sigma < \sigma_0} \|\partial_t^2 \log a_\sigma\|_{L^\infty(I)} < \infty. \quad (\text{C.60})$$

Then, uniformly for $w \in \mathcal{W}_{w_}$, all sufficiently small $\sigma > 0$ give exactly m nondegenerate maxima and $m - 1$ nondegenerate minima of*

$$F_{\sigma,w}(t) = a_\sigma(t)H_{\sigma,w}(x(t)) \quad (\text{C.61})$$

on I , ordered alternately, with no stationary endpoint. Relative to the pure roots in the x coordinate,

$$x(t_{j,\sigma}^{\max}) - p_j(\sigma, w) = O(\sigma^2), \quad (\text{C.62})$$

$$x(t_{j,\sigma}^{\min}) - r_j(\sigma, w) = O(\sigma^4), \quad (\text{C.63})$$

uniformly in w . Thus

$$x(t_{j,\sigma}^{\min}) = s_j(\sigma, w) + O(\sigma^4). \quad (\text{C.64})$$

Proof. Put

$$b_\sigma = \partial_t \log a_\sigma, \quad D_{\sigma,w}(t) = \partial_t \log F_{\sigma,w}(t) = b_\sigma(t) + x'(t)L_{\sigma,w}(x(t)). \quad (\text{C.65})$$

Since $F_{\sigma,w} > 0$, its stationary points are precisely the zeros of $D_{\sigma,w}$, and

$$D'_{\sigma,w} = b'_\sigma + x''L_{\sigma,w} + (x')^2 L'_{\sigma,w}. \quad (\text{C.66})$$

Choose $K > (M_0 + 1)/v_0$ and apply lemma C.6. On the full complement of the open boxes, the term $x'L$ strictly dominates b_σ , so D has the alternating signs in Eq. (C.54) and has no complement zero.

On a peak box, Eqs. (C.50) and (C.66) give

$$D'_{\sigma,w} \leq M_1 + \|x''\|_\infty C_p - v_0^2 \kappa_p \sigma^{-2} < 0 \quad (\text{C.67})$$

for small σ . Its boundary signs give exactly one zero. At that zero $F'' = FD' < 0$, hence the root is a nondegenerate maximum. Similarly, on a valley box,

$$D'_{\sigma,w} \geq -M_1 - \|x''\|_\infty C_v + v_0^2 \kappa_v \sigma^{-4} > 0, \quad (\text{C.68})$$

so its unique zero is a nondegenerate minimum. The certified tail signs exclude endpoint stationary points.

At a slow-factor root, Eq. (C.65) gives $|L| \leq M_0/v_0$. The pure peak root has $L(p_j) = 0$, and the uniform negative $L' = \Theta(\sigma^{-2})$ on its box implies the peak displacement in Eq. (C.62). The uniform positive $L' = \Theta(\sigma^{-4})$ on a valley box gives Eq. (C.63). Combining the latter with Eq. (C.38) proves Eq. (C.64). Every constant is uniform over the compact allocation family. $\square$

C.4 Fixed- ε Doi transfer and theorem boundary

For $B > 0$, let

$$f_{B,\varepsilon,w}(t) = \mathbb{E}_0 \left[K_{B,w,\varepsilon}(Z_t, R_t) \exp \left(- \int_0^t K_{B,w,\varepsilon}(Z_s, R_s) ds \right) \right] \quad (\text{C.69})$$

be the Doi reaction-time density and set

$$\mathcal{F}_{B,\varepsilon,w} = f_{B,\varepsilon,w}/B. \quad (\text{C.70})$$

This is budget-rescaling, not normalization to unit time integral.

Theorem C.8 (Exact m Doi modes for fixed finite (d, m)). *Fix all data in Eqs. (C.1)–(C.21), including the stationary midpoint variance, mutual independence, the strict weighted-space variance inequalities, target times strictly inside I , consistently oriented physical centres, normalized slabs of physical width $\varepsilon\rho$, the whole-window contact margin, and the compact allocation family. Then there is $\varepsilon_0 > 0$, depending on these fixed data, such that for every $0 < \varepsilon < \varepsilon_0$ and every $w \in \mathcal{W}_{w*}$, the continuum free-exposure clock $G_{\varepsilon,w}$ has exactly m nondegenerate maxima and $m - 1$ nondegenerate minima on I , ordered alternately, and no stationary endpoint.*

For each fixed such ε , there is $B_0(\varepsilon) > 0$, uniform over $w \in \mathcal{W}_{w}$, such that $\mathcal{F}_{B,\varepsilon,w}$ has the same complete stationary signature for every $0 < B < B_0(\varepsilon)$. Multiplication by $B > 0$ changes no stationary point, so the statement also holds for the physical density $f_{B,\varepsilon,w}$. This is theorem 1 of section 4, with its hypotheses written out in full.*

Proof. In lemma C.2, take $a_\sigma = a_\varepsilon$, with $\sigma = \varepsilon S_*/\ell_0$. Lemma C.1 supplies Eq. (C.60); hence theorem C.7 proves the exact stationary signature of $G_{\varepsilon,w}$, uniformly over the compact allocation family.

Fix one $0 < \varepsilon < \varepsilon_0$. The ordered simple roots depend continuously on w by the implicit-function theorem. Their finitely many graphs over the compact set $\mathcal{W}_{w*}$ are compact and disjoint. Continuity of the second time derivative gives pairwise disjoint root tubes on which G'' has a uniform nonzero sign and magnitude. After shrinking the tubes if necessary, G' has fixed opposite signs at the two boundaries of each tube. On the compact complement in $I \times \mathcal{W}_{w*}$, $|G'|$ has a positive minimum; the two endpoint slopes have positive uniform absolute minima as well.

For this fixed ε , $V_{w,\varepsilon}$ is bounded uniformly in w , and Eqs. (C.4)–(C.6) put q_0 in the required weighted state space. Because $w_j \geq w_* > 0$, the compact allocation set lies strictly inside the simplex and admits the affine control chart and a complex tube required by theorem B.1. (If the set is the singleton $w_j = 1/m$, the same conclusion is the corresponding pointwise case.) Applying that theorem with time orders 0, 1, 2 and no control derivative gives

$$\sup_{w \in \mathcal{W}_{w*}} \|\mathcal{F}_{B,\varepsilon,w} - G_{\varepsilon,w}\|_{C^2(I)} \longrightarrow 0 \quad (B \downarrow 0). \quad (\text{C.71})$$

Choose $B_0(\varepsilon)$ so that the C^1 error is smaller than all boundary, complement, and endpoint margins and the second-derivative error is smaller than the root-tube curvature margin. In each tube, $\mathcal{F}'$ has opposite boundary signs and $\mathcal{F}''$ keeps the sign of G'' , so it has exactly one root of the same type. The complement and endpoints contain none. This proves the complete finite-window signature. $\square$

Remark C.9 (Quantifiers, saturation, and scope). The quantifier order is

$$\text{fix } (d, m, \text{all geometric and probabilistic data}) \implies 0 < \varepsilon < \varepsilon_0 \implies 0 < B < B_0(\varepsilon). \quad (\text{C.72})$$

There is no asserted lower bound for $B_0(\varepsilon)$ uniform as $\varepsilon \downarrow 0$, and no interchange of the two limits. The proof permits $B_0(\varepsilon)$ to shrink rapidly as ε decreases—potentially exponentially in $1/\varepsilon^2$ —and no rate is claimed; this is why no common useful budget is asserted across the family. The construction is not uniform in d or m , does not provide one geometry with arbitrarily many modes, does not count stationary points outside I , and gives no process-level event-mass floor. It treats normalized longitudinal Gaussian slabs, not arbitrary localized catalyst patches.

Moreover, lemma C.1 shows that the relative contact factor is asymptotically saturated on the theorem window:

$$c_{d,\varepsilon} = 1 + O(\varepsilon^{-N} e^{-q/\varepsilon^2}) \quad \text{in } C^2(I). \quad (\text{C.73})$$

Thus the exact Doi embedding is rigorous, but the modal basis is created by the ordered narrow slabs sampled by the monotone constant-variance midpoint; the relative encounter factor is a slow perturbation in this asymptotic family. Nontrivial contact, a useful common positive budget, deterministic finite-parameter complement certificates, survival, and event-basin masses remain separate continuum-validation requirements rather than consequences of theorem C.8.

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